Study Guide: Lectures 1–3

Probability $\longrightarrow$ Inference $\longrightarrow$ Regression

25117 – Econometrics Universitat Pompeu Fabra

- **Lecture 1:** describe distributions and dependence; understand random sampling, estimators, and their sampling distributions.
- **Lecture 2:** quantify uncertainty with standard errors, tests, confidence intervals, and comparisons of means.
- **Lecture 3:** derive and interpret ordinary least squares, assess fit, and distinguish statistical association from causality.

How to study

For each formula, identify the object being estimated, the meaning of each symbol, the assumptions, and the interpretation. Work the examples before checking the answers.

Parameters, Random Variables, Estimators, and Estimates

Object	Notation	What is random?
Population mean	$\mu_Y = \mathbb{E}[Y]$	Fixed, usually unknown
Random sample	$Y_1, \ldots, Y_n$	Values vary across samples
Observed sample	$y_1, \ldots, y_n$	Recorded numbers
Estimator of the mean	$\overline{Y} = n^{-1} \sum_i Y_i$	A function of random data
Estimate of the mean	$\bar{y} = n^{-1} \sum_i y_i$	One realized number

- Capitals denote random variables; lower-case letters denote their realizations.
- Example: observing $(y_1, y_2, y_3) = (170, 180, 175)$ gives $\bar{y} = 175$ cm.
- The **sampling distribution** describes the different values of $\overline{Y}$ across repeated samples of the same size.

Unknown does not mean random: in frequentist inference, μ_Y is fixed even when we do not know it.

Probability Distributions: Mass, Density, and Cumulative Probability

A random variable assigns a number to each outcome of an experiment. An **event** is a set of outcomes.

Discrete: probability mass function

$$p_Y(y) = \Pr(Y = y), \qquad \sum_y p_Y(y) = 1.$$

Example: for a Bernoulli variable,

$$\Pr(Y = 1) = p, \qquad \Pr(Y = 0) = 1 - p.$$

Each possible value can have positive probability.

Continuous: probability density function

$$f_Y(y) \geq 0, \qquad \int_{-\infty}^{\infty} f_Y(y)\, dy = 1.$$

$$\Pr(a < Y \leq b) = \int_a^b f_Y(y)\, dy.$$

Probability is **area**, not height; $\Pr(Y = y) = 0$ at any single point.

For either type, the **cumulative distribution function (CDF)** is

$$F_Y(y) = \Pr(Y \leq y), \qquad \Pr(a < Y \leq b) = F_Y(b) - F_Y(a).$$

Essential Distinction: Standard Deviation and Standard Error

For independent and identically distributed observations with variance σ_Y^2:

Quantity	Formula	Interpretation
Population standard deviation	$\sigma_Y = \sqrt{\text{Var}(Y_i)}$	Spread of individual observations
SD of the sample mean	$\sigma_{\overline{Y}} = \sigma_Y/\sqrt{n}$	Spread of means across repeated samples
Sample standard deviation	$s_Y = \sqrt{\frac{\sum_i (y_i - \overline{y})^2}{n-1}}$	Estimate of σ_Y from one sample
Estimated standard error	$SE(\overline{Y}) = \frac{s_Y}{\sqrt{n}}$	Estimated sampling uncertainty of the mean

When do we replace σ_Y with s_Y?

When σ_Y is unknown, as it usually is. We still divide by $\sqrt{n}$: s_Y alone is *not* the standard error of $\overline{Y}$.

Example: if $n = 100$ and $s_Y = 20$ cm, then $SE(\overline{Y}) = 2$ cm. More observations make the mean more precise; they need not make individuals more similar.

Moments: Mean, Variance, and Standard Deviation

$$\mu_Y = \mathbb{E}[Y] = \sum_y y\, p_Y(y) \quad \text{or} \quad \int_{-\infty}^{\infty} y f_Y(y)\, dy,$$
$$\sigma_Y^2 = \mathrm{Var}(Y) = \mathbb{E}[(Y-\mu_Y)^2] = \mathbb{E}[Y^2] - \mu_Y^2,$$
$$\sigma_Y = \sqrt{\sigma_Y^2}.$$

- The mean describes location; variance and standard deviation describe dispersion.
- Standard deviation has the units of Y; variance has squared units.
- For constants a, b: $\mathbb{E}[a+bY] = a + b\mathbb{E}[Y]$ and $\mathrm{Var}(a+bY) = b^2\,\mathrm{Var}(Y)$.

Worked example

If Y takes $1, 2, 4$ with probabilities $.25, .50, .25$, then $\mathbb{E}[Y] = 2.25$, $\mathbb{E}[Y^2] = 6.25$, and $\mathrm{Var}(Y) = 6.25 - 2.25^2 = 1.1875$.

Joint and Marginal Distributions: Read the Whole Table

Let $X = 1$ mean a sunny day and $Y = 1$ mean buying ice cream.

	$X = 0$	$X = 1$	Marginal of Y
$Y = 0$	.15	.25	.40
$Y = 1$	.05	.55	.60
Marginal of X	.20	.80	1

- A **joint probability** concerns both events: $\Pr(X = 1, Y = 1) = .55$.
- A **marginal probability** ignores the other variable by summing over it:

$$p_Y(y) = \sum_x p_{XY}(x, y), \qquad p_X(x) = \sum_y p_{XY}(x, y).$$

- Thus $\Pr(Y = 1) = .05 + .55 = .60$ and $\Pr(X = 1) = .25 + .55 = .80$.

All joint cells are nonnegative and sum to 1. For continuous variables, integrate the joint density instead of summing probabilities.

Conditional Distributions: Change the Reference Group

For $\Pr(X = x) > 0$,

$$\Pr(Y = y \mid X = x) = \frac{\Pr(X = x, Y = y)}{\Pr(X = x)}.$$

1. Restrict attention to outcomes with $X = x$.
2. Divide each relevant joint cell by the probability of that group.
3. Check that the conditional probabilities sum to 1.

$$\Pr(Y = 1 \mid X = 1) = .55/.80 = .6875,$$
$$\Pr(Y = 0 \mid X = 1) = .25/.80 = .3125.$$

Reversing the condition changes the denominator:

$$\Pr(X = 1 \mid Y = 1) = .55/.60 \approx .9167.$$

Bayes' rule: $\Pr(X = x \mid Y = y) = \Pr(Y = y \mid X = x) \Pr(X = x) / \Pr(Y = y)$.

Conditional Means and the Law of Iterated Expectations

The conditional mean averages Y **within** a group:

$$m(x) = \mathbb{E}[Y \mid X = x] = \sum_y y \Pr(Y = y \mid X = x).$$

For binary Y, $\mathbb{E}[Y \mid X = x] = \Pr(Y = 1 \mid X = x)$. In the sunny-day example,

$$m(0) = .05/.20 = .25, \qquad m(1) = .55/.80 = .6875.$$

The **Law of Iterated Expectations (LIE)** averages the group means using their population shares:

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[\mathbb{E}[Y \mid X]] \\ &= \sum_x \mathbb{E}[Y \mid X = x] \Pr(X = x) \\ &= .25(.20) + .6875(.80) = .60.\end{aligned}$$

What is random? $m(x)$ is a number at a specified x; $m(X) = \mathbb{E}[Y \mid X]$ is a function of the random variable X. Do not give all groups equal weight unless they have equal probability.

Conditional-Expectation Rules You Will Use in Regression

1. **Linearity:** $\mathbb{E}[aY + bZ \mid X] = a\mathbb{E}[Y \mid X] + b\mathbb{E}[Z \mid X]$ for constants a, b.
2. **Take out what is known:**
$$\mathbb{E}[g(X)Y \mid X] = g(X)\mathbb{E}[Y \mid X].$$
Why? Once $X = x$ is fixed, $g(x)$ is a constant:
$$\mathbb{E}[g(X)Y \mid X = x] = g(x)\mathbb{E}[Y \mid X = x].$$
In particular, $\mathbb{E}[XY \mid X] = X\mathbb{E}[Y \mid X]$.
3. **Conditional variance:**
$$\mathrm{Var}(Y \mid X = x) = \mathbb{E}[(Y - m(x))^2 \mid X = x].$$

A useful implication combines the first two rules with iterated expectations:

$$\mathbb{E}[u \mid X] = 0 \implies \mathbb{E}[u] = 0, \quad \mathbb{E}[Xu] = \mathbb{E}[X\mathbb{E}[u \mid X]] = 0.$$

Hence $\mathrm{Cov}(X, u) = 0$.

Independence, Covariance, and Correlation

Independence: knowing X does not change the distribution of Y:

$$p_{XY}(x, y) = p_X(x)p_Y(y) \quad \iff \quad p_{Y|X}(y \mid x) = p_Y(y).$$

Covariance measures linear comovement; **correlation** removes the units:

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \qquad \rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

- Independence implies zero covariance (with finite moments).
- Zero covariance does **not** imply independence. If X is equally likely to be $-1, 0, 1$ and $Y = X^2$, then $\mathrm{Cov}(X, Y) = 0$ but Y depends on X.
- In the sunny-day table: $\mathrm{Cov}(X, Y) = .55 - (.80)(.60) = .07 > 0$. The variables are not independent.

$$\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y).$$

Random Sampling and the Distribution of the Sample Mean

An **independent and identically distributed (i.i.d.) sample** has:

- The same population distribution for every Y_i.
- Independence across observations. This does not require X_i and Y_i within the same observation to be independent.

For i.i.d. observations with finite variance,

$$\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i,$$

$$\mathbb{E}[\overline{Y}] = \frac{1}{n}\sum_{i} \mathbb{E}[Y_i] = \mu_Y,$$

$$\mathrm{Var}(\overline{Y}) = \frac{1}{n^2}\sum_{i} \mathrm{Var}(Y_i) = \frac{n\sigma_Y^2}{n^2} = \frac{\sigma_Y^2}{n}.$$

The covariance terms disappear because the observations are independent. Thus $SD(\overline{Y}) = \sigma_Y/\sqrt{n}$: quadrupling n halves this standard deviation.

Law of Large Numbers: Why the Mean Is Consistent

The **Law of Large Numbers (LLN)** describes convergence of the sample mean. For i.i.d. observations with $\mathbb{E}[|Y_i|] < \infty$,

$$\overline{Y} \xrightarrow{p} \mu_Y.$$

Convergence in probability means that, for every tolerance $\varepsilon > 0$,

$$\Pr(|\overline{Y} - \mu_Y| > \varepsilon) \longrightarrow 0.$$

- As the sample grows, a large estimation error becomes unlikely.
- This does **not** mean $\Pr(\overline{Y} = \mu_Y) \to 1$. Exact equality need never occur for continuous data.
- It does not mean that every additional observation moves the estimate closer to μ_Y.

Example

If Y_i indicates a head on an independent fair coin toss, $\overline{Y}$ is the sample fraction of heads and converges in probability to 0.5.

Central Limit Theorem: Why Normal Approximations Work

The **Central Limit Theorem (CLT)** describes the shape of sampling uncertainty. For i.i.d. observations with $0 < \sigma_Y^2 < \infty$,

$$\frac{\overline{Y} - \mu_Y}{\sigma_Y/\sqrt{n}} = \frac{\sqrt{n}(\overline{Y} - \mu_Y)}{\sigma_Y} \xrightarrow{d} N(0,1).$$

1. **Center:** subtract μ_Y to measure the estimation error.
2. **Scale:** divide by the standard deviation of the mean, $\sigma_Y/\sqrt{n}$.
3. **Approximate:** for sufficiently large n, compare this standardized error with a standard normal distribution.

Equivalently, $\overline{Y} \approx N(\mu_Y, \sigma_Y^2/n)$ in large samples.

LLN versus CLT

The LLN says the error becomes small. The CLT describes its approximate distribution and scale. It does not say that individual observations become normal.

Estimator Properties: Unbiasedness, Consistency, and Efficiency

Property	Definition	Question it answers
Unbiasedness	$\mathbb{E}[\widehat{\theta}] = \theta$	Is the estimator centered on the truth across samples?
Consistency	$\widehat{\theta} \xrightarrow{p} \theta$	Does the estimation error vanish as n grows?
Efficiency	Smallest variance in a specified class	Which eligible estimator is most precise?

- For i.i.d. data with finite variance, $\overline{Y}$ is unbiased and consistent.
- Unbiasedness alone does not imply consistency: using Y_1 to estimate μ_Y is unbiased but its variance does not shrink with n.
- When comparing biased estimators, variance alone is insufficient:

$$MSE(\widehat{\theta}) = \text{Var}(\widehat{\theta}) + \text{Bias}(\widehat{\theta})^2.$$

Here $\text{Bias}(\widehat{\theta}) = \mathbb{E}[\widehat{\theta}] - \theta$.

BLUE, Least Squares, and the Cramér–Rao Bound

BLUE means **Best Linear Unbiased Estimator**. For i.i.d. observations with common finite variance, a linear estimator $\sum_i w_i Y_i$ is unbiased for every μ_Y if $\sum_i w_i = 1$.

$$\mathrm{Var}\left(\sum_i w_i Y_i\right) = \sigma_Y^2 \sum_i w_i^2 \geq \frac{\sigma_Y^2}{n}.$$

Equal weights $w_i = 1/n$ attain the bound: $\overline{Y}$ is BLUE.

Least squares gives the same estimator by a different argument:

$$\min_m \sum_i (Y_i - m)^2 \quad \Longrightarrow \quad -2\sum_i (Y_i - m) = 0 \quad \Longrightarrow \quad m = \overline{Y}.$$

Cramér–Rao: under regularity conditions, an unbiased estimator satisfies $\mathrm{Var}(\widehat{\mu}_Y) \geq 1/\mathcal{I}_n(\mu_Y)$. For normal observations with known variance, the sample's **Fisher information** is $\mathcal{I}_n(\mu_Y) = n/\sigma_Y^2$. The mean reaches this bound even among nonlinear unbiased estimators.

BLUE concerns a finite-sample comparison class; consistency is a separate large-sample property. Normality is not needed for the BLUE result above.

Estimating an Unknown Variance and the Mean's Standard Error

Before observing the sample, write the random sample variance as

$$S_Y^2 = \frac{1}{n-1}\sum_i (Y_i - \overline{Y})^2.$$

After observing it, compute the number

$$s_Y^2 = \frac{1}{n-1}\sum_i (y_i - \overline{y})^2, \qquad s_Y = \sqrt{s_Y^2}.$$

- We estimate μ_Y with $\overline{y}$. Since $\sum_i (y_i - \overline{y}) = 0$, only $n-1$ deviations can vary freely.
- Division by $n-1$ makes S_Y^2 unbiased for σ_Y^2 under i.i.d. sampling. Its square root need not be unbiased for σ_Y.
- When σ_Y is unknown, estimate the mean's SD by

$$\boxed{SE(\overline{Y}) = \frac{s_Y}{\sqrt{n}}}.$$

This guide writes S_Y for the random statistic and s_Y for its observed value. The lectures often use s_Y for both; the distinction is the same as $\overline{Y}$ versus $\overline{y}$.

Known σ_Y or Estimated s_Y: Which Reference Distribution?

Situation	Random test statistic under $H_0 : \mu_Y = \mu_0$	Reference distribution
Known σ_Y	$Z = \dfrac{\overline{Y} - \mu_0}{\sigma_Y/\sqrt{n}}$	Exactly $N(0,1)$ for i.i.d. normal data; approximately normal for large n by the CLT.
Unknown σ_Y	$T = \dfrac{\overline{Y} - \mu_0}{S_Y/\sqrt{n}}$	Exactly Student's t_{n-1} for i.i.d. normal data; approximately $N(0,1)$ for large n under the CLT conditions.

- The observed number is $t_{\text{obs}} = (\overline{y} - \mu_0)/(s_Y/\sqrt{n})$.
- Estimating σ_Y adds uncertainty: the finite-sample t distribution has heavier tails.
- In large samples, $S_Y/\sigma_Y \xrightarrow{p} 1$, so replacing σ_Y does not change the limiting standard normal distribution.

For small, non-normal samples, neither an exact t claim nor an automatic normal approximation is justified.

Hypothesis Testing: A Procedure, Not Just a Formula

1. **State the parameter and null:** for example, $H_0 : \mu_Y = 175$ cm.
2. **Choose the alternative:** $H_1 : \mu_Y \neq 175$ is two-sided; $H_1 : \mu_Y > 175$ is one-sided.
3. **Choose a significance level** α before looking at the result.
4. **Compute the standard error** and the observed statistic:

$$t_{\text{obs}} = \frac{\overline{y} - \mu_0}{SE(\overline{Y})}.$$

5. **Use the appropriate null distribution** to calculate a p-value.
6. **Decide and interpret:** reject if $p \leq \alpha$; otherwise do not reject.

Interpret the statistic

A value $t_{\text{obs}} = -2.4$ means the observed mean is 2.4 estimated standard errors below the null value. It is not a probability.

p-Values, Tail Areas, and Significance Levels

A p-**value is a probability under the null hypothesis**: how often would the test statistic be at least as extreme as the observed one? For $Z \sim N(0,1)$, define $\Phi(z) = \Pr(Z \leq z)$.

Alternative	Normal-approximation p-value	Reject at 5% if
$\mu_Y \neq \mu_0$	$2[1 - \Phi(\lvert t_{\text{obs}} \rvert)]$	$\lvert t_{\text{obs}} \rvert \geq 1.96$

For two-sided normal tests, critical values are:

$$\alpha = 10\% : \ 1.645, \qquad \alpha = 5\% : \ 1.960, \qquad \alpha = 1\% : \ 2.576.$$

Use Student's t probabilities and critical values when that is the reference distribution.

Avoid these interpretations: the p-value is not $\Pr(H_0 \text{ is true})$; failure to reject does not establish that H_0 is true. Statistical significance does not measure economic importance.

Worked Example: Height, Standard Error, and a Mean Test

The ten illustrative heights from Lecture 2 have

$$n = 10, \quad \bar{y} = 170 \text{ cm}, \quad \sum_i (y_i - \bar{y})^2 = 388 \text{ cm}^2.$$

$$\begin{aligned} s_Y &= \sqrt{388/9} = 6.5659 \text{ cm}, \\ SE(\overline{Y}) &= 6.5659/\sqrt{10} = 2.0763 \text{ cm}, \\ t_{\text{obs}} &= \frac{170 - 175}{2.0763} = -2.4081. \end{aligned}$$

For $H_0 : \mu_Y = 175$ versus $H_1 : \mu_Y \neq 175$:

- **Normal approximation:** $p \approx .0160$ and $|t_{\text{obs}}| > 1.96$.
- **If heights are i.i.d. normal and σ_Y is unknown:** use t_9; $p \approx .0394$ and the 5% critical value is 2.262.
- Both reject at 5%, but their p-values differ. With only ten observations, normality is an additional assumption, not a consequence of a large sample.

Do not divide by s_Y alone: that describes individual dispersion, not uncertainty in the mean.

Confidence Intervals: Formula, Meaning, and Link to Tests

A two-sided $(1-\alpha)$ confidence interval for a mean is

$$\bar{y} \pm c\,SE(\overline{Y}),$$

where $c = z_{1-\alpha/2}$ for a normal approximation, or $c = t_{1-\alpha/2,n-1}$ for exact inference with i.i.d. normal data and unknown variance.

- **Coverage:** over repeated samples, the interval procedure contains the fixed true parameter in $100(1-\alpha)\%$ of samples (approximately for asymptotic intervals).
- **Width:** more confidence gives a wider interval; greater precision gives a narrower interval.
- **Test equivalence:** a null value outside the interval is rejected by the corresponding two-sided test, using the same standard error and reference distribution.

For the height example:

$$\text{Normal 95\% CI: } 170 \pm 1.96(2.0763) \approx [165.93, 174.07],$$

$$\text{Normal-population } t_9 \text{ CI: } 170 \pm 2.262(2.0763) \approx [165.30, 174.70].$$

These are intervals for the **population mean**, not ranges containing 95% of individual heights.

Type I Error, Type II Error, and Power

	H_0 true	A particular alternative true
Reject H_0	Type I error	Correct rejection
Do not reject H_0	Correct non-rejection	Type II error

For a test calibrated to have size α,

$$\alpha = \mathsf{Pr}(\text{reject } H_0 \mid H_0 \text{ true}).$$

At a specified alternative mean μ_1,

$$\beta(\mu_1) = \mathsf{Pr}_{\mu_1}(\text{do not reject } H_0), \qquad \mathsf{Power}(\mu_1) = 1 - \beta(\mu_1).$$

- α controls false rejection under the null. β depends on the particular true alternative, sample size, and noise.
- For a fixed sample size and test family, lowering α makes rejection harder and generally reduces power.
- Larger samples or effects farther from the null generally increase power.

How to Calculate β: Use a Specific Alternative

Consider a two-sided normal mean test with known σ_Y and rejection rule

$$|\overline{Y} - \mu_0| > c\,\frac{\sigma_Y}{\sqrt{n}}, \qquad c = z_{1-\alpha/2}.$$

If the true mean is μ_1, define the effect in standard-error units:

$$\delta = \frac{\mu_1 - \mu_0}{\sigma_Y/\sqrt{n}}.$$

Under this alternative, the test statistic has distribution $N(\delta, 1)$, so

$$\boxed{\beta(\mu_1) = \Phi(c - \delta) - \Phi(-c - \delta)}.$$

Example

Let $\alpha = .05$, $c = 1.96$, and $\delta = 2$. Then $\beta = \Phi(-.04) - \Phi(-3.96) \approx .4840$, so power is about $.5160$.

The formula is exact for normal observations with known variance and approximate when justified by the CLT. There is no single β determined by α alone.

Comparing Two Means: Independent Groups

For independent samples from groups a and b, let $\Delta = \mu_a - \mu_b$ and test $H_0 : \Delta = d_0$.

$$\widehat{\Delta} = \overline{Y}^a - \overline{Y}^b,$$

$$\mathrm{Var}(\widehat{\Delta}) = \frac{\sigma_a^2}{n_a} + \frac{\sigma_b^2}{n_b},$$

$$SE(\widehat{\Delta}) = \sqrt{\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}},$$

$$t_{\mathrm{obs}} = \frac{(\overline{y}^a - \overline{y}^b) - d_0}{SE(\widehat{\Delta})}.$$

- Subtract the hypothesized difference d_0, which need not be zero.
- Large-sample 95% CI: $(\overline{y}^a - \overline{y}^b) \pm 1.96\, SE(\widehat{\Delta})$.
- Independence removes the covariance term. For paired measurements on the same people, analyze within-person differences instead.
- For small independent normal samples with unknown unequal variances, use Welch's t method rather than treating 1.96 as exact.

Group Differences: Calculation and Causal Interpretation

From the two height samples in Lecture 2:

$$n_a = n_b = 10, \quad \bar{y}^a = 170, \quad \bar{y}^b = 164, \quad s_a^2 = 388/9, \quad s_b^2 = 132/9.$$

$$SE(\widehat{\Delta}) = \sqrt{\frac{388/9}{10} + \frac{132/9}{10}} = 2.4037.$$

For $H_0 : \Delta = 5$ cm:

$$t_{\text{obs}} = \frac{(170 - 164) - 5}{2.4037} = .4160, \qquad p \approx .6774$$

using the lecture's rough normal approximation. Do not reject at 5%.

A difference in conditional means is initially an association:

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0].$$

Random assignment, together with a well-defined treatment, no spillovers, and appropriate observation of outcomes, can justify a causal interpretation. In observational data, different group composition can explain part or all of the difference.

Simple Linear Regression: Three Different Objects

Object	Equation	Meaning
Population model	$Y_i = \beta_0 + \beta_1 X_i + u_i$	Unknown parameters and unobserved error
Estimated rule before observing data	$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i$	Random estimators and fitted values
Fitted line in an observed sample	$\widehat{y}_i = \widehat{b}_0 + \widehat{b}_1 x_i$	Numerical estimates and predictions

- β_0 is an intercept; β_1 is a change in Y per unit of X.
- u_i contains determinants of Y_i omitted from the displayed part of the model.
- A sample residual $\widehat{u}_i$ is not the same object as the population error u_i.

For clarity, this guide uses $\widehat{b}_j$ for a realized estimate of $\widehat{\beta}_j$; the lectures often retain $\widehat{\beta}_j$ in numerical applications.

Ordinary Least Squares: Objective and First-Order Conditions

Ordinary least squares (OLS) chooses an intercept a and slope b to minimize

$$Q_n(a, b) = \sum_{i=1}^{n} (Y_i - a - bX_i)^2.$$

The **first-order conditions (FOCs)** are

$$\frac{\partial Q_n}{\partial a} = -2 \sum_i (Y_i - a - bX_i) = 0,$$

$$\frac{\partial Q_n}{\partial b} = -2 \sum_i X_i (Y_i - a - bX_i) = 0.$$

Evaluate them at $(a, b) = (\widehat{\beta}_0, \widehat{\beta}_1)$. The intercept condition gives

$$n\overline{Y} - n\widehat{\beta}_0 - n\widehat{\beta}_1 \overline{X} = 0 \quad \Longrightarrow \quad \boxed{\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X}}.$$

If $TSS_X = \sum_i (X_i - \overline{X})^2 > 0$, the squared-error objective has a unique minimizer.

Solve the Slope FOC: Center, Substitute, and Divide

Substitute the intercept expression into the slope FOC:

$$\begin{aligned} 0 &= \sum_i X_i[Y_i - (\overline{Y} - \widehat{\beta}_1 \overline{X}) - \widehat{\beta}_1 X_i] \\ &= \sum_i X_i[(Y_i - \overline{Y}) - \widehat{\beta}_1 (X_i - \overline{X})]. \end{aligned}$$

Since $\sum_i (Y_i - \overline{Y}) = 0$ and $\sum_i (X_i - \overline{X}) = 0$,

$$\sum_i X_i (Y_i - \overline{Y}) = \sum_i (X_i - \overline{X})(Y_i - \overline{Y}),$$

$$\sum_i X_i (X_i - \overline{X}) = \sum_i (X_i - \overline{X})^2 = TSS_X.$$

TSS_X is the **Total Sum of Squares of** X: the total variation of the sampled X_i values around their sample mean. Therefore,

$$\boxed{\widehat{\beta}_1 = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{TSS_X} = \frac{S_{XY}}{S_X^2}}.$$

Here S_{XY} and S_X^2 denote the random sample covariance and variance, each dividing its sum by $n - 1$. Their observed values are s_{XY} and s_X^2.

Worked OLS Example: From Sums to a Fitted Line

For the ten observations in Lecture 3, the observed summaries are

$$n = 10, \quad \sum_i x_i = 55, \quad \sum_i y_i = 84.90, \quad \bar{x} = 5.5, \quad \bar{y} = 8.49,$$

$$\sum_i (x_i - \bar{x})^2 = 82.50, \qquad \sum_i (x_i - \bar{x})(y_i - \bar{y}) = -31.75.$$

1. Compute the sample variance and covariance: $s_X^2 = 82.50/9 = 9.1667$, $s_{XY} = -31.75/9 = -3.5278$.
2. Divide to obtain the slope: $\widehat{b}_1 = -31.75/82.50 = -.38485$.
3. Recover the intercept: $\widehat{b}_0 = 8.49 - (-.38485)(5.5) = 10.60667$.

Thus $\widehat{y}_i = 10.60667 - .38485 x_i$. At $x_1 = 1$, $\widehat{y}_1 = 10.22182$; since $y_1 = 10.8$, the residual is $.57818$.

A one-unit increase in X is associated with about $.385$ fewer units of Y. This slope is not automatically causal.

OLS Identities: What Is Zero by Construction?

For $Q_n(a, b) = \sum_i (Y_i - a - bX_i)^2$, evaluate the first-order conditions at $(a, b) = (\widehat{\beta}_0, \widehat{\beta}_1)$:

$$\left.\frac{\partial Q_n}{\partial a}\right|_{\widehat{\beta}_0, \widehat{\beta}_1} = -2 \sum_i \widehat{u}_i = 0 \quad \Longrightarrow \quad \sum_i \widehat{u}_i = 0,$$

$$\left.\frac{\partial Q_n}{\partial b}\right|_{\widehat{\beta}_0, \widehat{\beta}_1} = -2 \sum_i X_i \widehat{u}_i = 0 \quad \Longrightarrow \quad \sum_i X_i \widehat{u}_i = 0.$$

The first is the intercept FOC; the second is the slope FOC.

- **Mean residual is zero:** $\overline{\widehat{u}} = 0$. Individual residuals are usually not zero.
- **The line passes through the means:** $\overline{Y} = \widehat{\beta}_0 + \widehat{\beta}_1 \overline{X}$.
- **Fitted values and residuals are orthogonal:**

$$\sum_i \widehat{Y}_i \widehat{u}_i = \widehat{\beta}_0 \sum_i \widehat{u}_i + \widehat{\beta}_1 \sum_i X_i \widehat{u}_i = 0.$$

An algebraic identity is not an identifying assumption

These identities hold for the fitted OLS residuals even when $\mathbb{E}[u_i \mid X_i] \neq 0$. A zero sample residual mean does not prove conditional mean zero for the population error.

Goodness of Fit: Three Sums of Squares and R^2

$$TSS_Y = \sum_i (Y_i - \overline{Y})^2 \qquad \text{total variation around the mean,}$$

$$ESS = \sum_i (\widehat{Y}_i - \overline{Y})^2 \qquad \text{variation explained by the fitted line,}$$

$$SSR = \sum_i (Y_i - \widehat{Y}_i)^2 \qquad \text{residual sum of squares.}$$

Because $Y_i - \overline{Y} = (\widehat{Y}_i - \overline{Y}) + \widehat{u}_i$ and the cross term sums to zero,

$$TSS_Y = ESS + SSR, \qquad \boxed{R^2 = \frac{ESS}{TSS_Y} = 1 - \frac{SSR}{TSS_Y}}.$$

- With an intercept and $TSS_Y > 0$, $0 \leq R^2 \leq 1$.
- $R^2 = .60$ means the fitted line accounts for 60% of the sample variation around the mean.
- High R^2 does not establish causality or guarantee good predictions in a new sample.

Keep TSS_X (variation in X) separate from TSS_Y (variation in Y).

SER, RMSE, and Coefficient Standard Errors Are Different

Standard error of the regression (SER)

$$\widehat{\sigma}_u^2 = \frac{SSR}{n-2} = SER^2, \qquad SER = \sqrt{\widehat{\sigma}_u^2}.$$

Two estimated coefficients use two degrees of freedom. Under the homoskedastic model, SER^2 estimates σ_u^2.

Root mean squared error (RMSE)

$$RMSE = \sqrt{\frac{SSR}{n}}.$$

The square root of the average squared in-sample prediction error: $Y_i - \widehat{Y}_i = \widehat{u}_i$.

- Both measure unexplained outcome dispersion, in units of Y; they become similar when n is large.
- $SE(\widehat{\beta}_1)$ instead measures uncertainty about the estimated slope, in units of Y per unit of X.
- Neither SER nor $\widehat{Y} \pm SER$ is by itself a calibrated 95% prediction interval.

Always check the denominator: some software labels $\sqrt{SSR/(n-2)}$ as Root MSE.

Estimating the Error Variance: Why Residuals Need a Correction

Assumptions. Condition on all sampled regressors $\mathbf{X}$: $\mathbb{E}[u_i \mid \mathbf{X}] = 0$, $\mathrm{Var}(u_i \mid \mathbf{X}) = \sigma_u^2$, and errors are uncorrelated conditional on $\mathbf{X}$; $TSS_X > 0$ and $n > 2$.

If the population coefficients were known

We could observe $u_i = Y_i - \beta_0 - \beta_1 X_i$. Since $\mathbb{E}[u_i^2 \mid \mathbf{X}] = \sigma_u^2$,

$$\mathbb{E}\left[\frac{1}{n}\sum_i u_i^2 \,\middle|\, \mathbf{X}\right] = \sigma_u^2.$$

Actual residuals use fitted coefficients: $\widehat{u}_i = Y_i - \widehat{\beta}_0 - \widehat{\beta}_1 X_i$. Least squares fits some of the realized error, so the residual sum of squares is smaller on average.

$$\mathbb{E}[SSR \mid \mathbf{X}] = (n-2)\sigma_u^2 \qquad \text{(derived on the next slide)},$$

$$\widehat{\sigma}_u^2 = \frac{SSR}{n-2}, \qquad \mathbb{E}[\widehat{\sigma}_u^2 \mid \mathbf{X}] = \sigma_u^2.$$

SER estimates σ_u; SER^2 is the unbiased variance estimator under these assumptions.

Why the Correction Is $n-2$: A Derivation

Under the previous assumptions, define $\overline{u} = n^{-1}\sum_i u_i$ and $d = \widehat{\beta}_1 - \beta_1$. The intercept identity gives

$$\widehat{\beta}_0 - \beta_0 = \overline{u} - d\overline{X} \quad \Longrightarrow \quad u_i = \widehat{u}_i + \overline{u} + d(X_i - \overline{X}).$$

OLS orthogonality, $\sum_i \widehat{u}_i = 0$ and $\sum_i (X_i - \overline{X})\widehat{u}_i = 0$, together with $\sum_i (X_i - \overline{X}) = 0$, eliminates all cross-products when squaring and summing:

$$SSR = \sum_i u_i^2 - n\overline{u}^2 - d^2\, TSS_X.$$

Using OLS unbiasedness and $\mathrm{Var}(\widehat{\beta}_1 \mid \mathbf{X}) = \sigma_u^2 / TSS_X$ **(results derived later in this guide)**, we obtain

$$\begin{aligned}
\mathbb{E}\left[\sum_i u_i^2 \,\middle|\, \mathbf{X}\right] &= n\sigma_u^2, \\
\mathbb{E}[n\overline{u}^2 \mid \mathbf{X}] &= n\,\mathrm{Var}(\overline{u} \mid \mathbf{X}) = \sigma_u^2, \\
\mathbb{E}[d^2\, TSS_X \mid \mathbf{X}] &= TSS_X\,\mathrm{Var}(\widehat{\beta}_1 \mid \mathbf{X}) = \sigma_u^2.
\end{aligned}$$

Thus $\mathbb{E}[SSR \mid \mathbf{X}] = n\sigma_u^2 - \sigma_u^2 - \sigma_u^2 = (n-2)\sigma_u^2$. One fitted direction for the intercept and one for the slope remove two σ_u^2 terms.

Population Least Squares and Sample OLS

The **population linear projection** minimizes an expected squared error:

$$(a^*, b^*) = \arg\min_{a,b} \mathbb{E}[(Y - a - bX)^2].$$

Its FOCs, when the moments exist, are

$$\mathbb{E}[Y - a^* - b^*X] = 0, \qquad \mathbb{E}[X(Y - a^* - b^*X)] = 0.$$

Thus, for $\mathrm{Var}(X) > 0$,

$$b^* = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}, \qquad a^* = \mathbb{E}[Y] - b^*\mathbb{E}[X].$$

- Population coefficients are fixed features of the joint distribution.
- Sample OLS replaces population moments with sample moments; the estimated coefficients vary across samples.
- If $\mathbb{E}[Y \mid X] = \beta_0 + \beta_1 X$, the projection equals that conditional-mean line.
- If the conditional mean is nonlinear, the projection remains the best *linear* predictor under squared loss.

Here a^*, b^* distinguish population projection coefficients from sample estimators and any structural causal parameters.

OLS Assumptions: What Each One Does

For a model $Y_i = \beta_0 + \beta_1 X_i + u_i$:

1. **Conditional mean zero:** $\mathbb{E}[u_i \mid X_i] = 0$. It implies $\mathbb{E}[Y_i \mid X_i] = \beta_0 + \beta_1 X_i$ and removes systematic error differences across values of X_i.
2. **i.i.d. observations:** the pairs (X_i, Y_i) are independent across i and drawn from one joint distribution. This supports the sampling arguments.
3. **Finite moments and regressor variation:** the lecture assumes finite fourth moments and $\mathrm{Var}(X) > 0$. These control tail behavior and ensure that a slope can be identified and estimated.

Causal interpretation needs a causal model

If β_1 is a structural causal effect, conditional mean zero helps identify it from observed variation. A fitted conditional-mean relationship alone does not prove that intervening on X changes Y.

Homoskedasticity, $\mathrm{Var}(u_i \mid X_i) = \sigma_u^2$, is an additional assumption for the simple variance formula; it is not required for OLS unbiasedness.

Omitted Variables: Why Conditional Mean Zero Can Fail

Suppose the structural equation is

$$Y_i = \beta_0 + \beta_1 X_i + \gamma Z_i + v_i, \qquad \mathbb{E}[v_i \mid X_i, Z_i] = 0.$$

If we omit Z_i and keep the displayed coefficients β_0, β_1, comparison with $Y_i = \beta_0 + \beta_1 X_i + u_i$ gives

$$u_i = \gamma Z_i + v_i, \qquad \mathbb{E}[u_i \mid X_i] = \gamma \mathbb{E}[Z_i \mid X_i].$$

If $\gamma \neq 0$ and the conditional mean of Z_i varies with X_i, the conditional mean of u_i varies too; an intercept shift cannot remove that variation.

The population projection slope becomes

$$b^* = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)} = \beta_1 + \gamma \frac{\mathrm{Cov}(X, Z)}{\mathrm{Var}(X)}.$$

For example, if Z is disadvantage, $\gamma < 0$, and $\mathrm{Cov}(X, Z) > 0$, omitting it shifts the slope downward. Conditional mean zero implies zero covariance; its failure does not always imply nonzero covariance.

Decompose the OLS Slope: True Coefficient Plus Sampling Error

Write $D_i = X_i - \overline{X}$ and $TSS_X = \sum_i D_i^2$. First,

$$\sum_i D_i(Y_i - \overline{Y}) = \sum_i D_i Y_i \quad \text{because} \quad \sum_i D_i = 0.$$

Substitute $Y_i = \beta_0 + \beta_1 X_i + u_i$:

$$\widehat{\beta}_1 = \frac{\beta_0 \sum_i D_i + \beta_1 \sum_i D_i X_i + \sum_i D_i u_i}{TSS_X}.$$

Now use $X_i = \overline{X} + D_i$:

$$\sum_i D_i X_i = \overline{X} \sum_i D_i + \sum_i D_i^2 = TSS_X.$$

Therefore,

$$\boxed{\widehat{\beta}_1 = \beta_1 + \frac{\sum_i (X_i - \overline{X}) u_i}{TSS_X}}.$$

The intercept term disappears, and the remaining fraction is the estimation error. This identity holds before taking expectations.

Why OLS Is Unbiased: Condition First, Then Average

Let $\mathbf{X} = (X_1, \ldots, X_n)$ denote **all** sampled regressors. Under i.i.d. sampling and conditional mean zero, $\mathbb{E}[u_i \mid \mathbf{X}] = 0$.

1. Conditional on $\mathbf{X}$, every X_i, $\overline{X}$, and TSS_X is fixed.
2. Take the conditional expectation of the decomposition:

$$\mathbb{E}[\widehat{\beta}_1 \mid \mathbf{X}] = \beta_1 + \frac{\sum_i (X_i - \overline{X})\mathbb{E}[u_i \mid \mathbf{X}]}{TSS_X} = \beta_1.$$

3. Apply the Law of Iterated Expectations:

$$\mathbb{E}[\widehat{\beta}_1] = \mathbb{E}[\mathbb{E}[\widehat{\beta}_1 \mid \mathbf{X}]] = \beta_1.$$

The next slide derives the corresponding result for the intercept.

These results require an identified slope ($TSS_X > 0$) and existence of the relevant expectations. Homoskedasticity and normality are not needed for this argument.

Why the Intercept Is Unbiased: Step by Step

Define $\overline{u} = \frac{1}{n}\sum_i u_i$.

1. Average the population equation:
$$\overline{Y} = \beta_0 + \beta_1 \overline{X} + \overline{u}.$$

2. Substitute into the intercept estimator:
$$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X} = \beta_0 + (\beta_1 - \widehat{\beta}_1)\overline{X} + \overline{u}.$$

3. Conditional on $\mathbf{X} = (X_1, \ldots, X_n)$, $\overline{X}$ is fixed and
$$\mathbb{E}[\overline{u} \mid \mathbf{X}] = \frac{1}{n}\sum_i \mathbb{E}[u_i \mid \mathbf{X}] = 0.$$

   Using the previous result $\mathbb{E}[\widehat{\beta}_1 \mid \mathbf{X}] = \beta_1$,
$$\begin{aligned}\mathbb{E}[\widehat{\beta}_0 \mid \mathbf{X}] &= \beta_0 + (\beta_1 - \mathbb{E}[\widehat{\beta}_1 \mid \mathbf{X}])\overline{X} + \mathbb{E}[\overline{u} \mid \mathbf{X}] \\ &= \beta_0 + (\beta_1 - \beta_1)\overline{X} + 0 = \beta_0.\end{aligned}$$

4. Apply iterated expectations:
$$\mathbb{E}[\widehat{\beta}_0] = \mathbb{E}[\mathbb{E}[\widehat{\beta}_0 \mid \mathbf{X}]] = \beta_0.$$

OLS Precision: Variance and Standard Error of the Slope

With errors uncorrelated across observations conditional on $\mathbf{X}$,

$$\begin{aligned}\operatorname{Var}(\widehat{\beta}_1 \mid \mathbf{X}) &= \frac{1}{TSS_X^2}\operatorname{Var}\left(\sum_i (X_i - \overline{X})u_i \mid \mathbf{X}\right) \\ &= \frac{\sum_i (X_i - \overline{X})^2 \operatorname{Var}(u_i \mid \mathbf{X})}{TSS_X^2}.\end{aligned}$$

Under **homoskedasticity**, every conditional error variance is σ_u^2, so

$$\operatorname{Var}(\widehat{\beta}_1 \mid \mathbf{X}) = \frac{\sigma_u^2}{TSS_X}, \qquad \boxed{SE(\widehat{\beta}_1) = \sqrt{\frac{SSR}{(n-2)\,TSS_X}} = \frac{SER}{\sqrt{TSS_X}}}.$$

- More error variation makes the slope less precise.
- More variation in X increases TSS_X and makes the slope more precise.
- With stable regressor variation, larger samples typically increase TSS_X.

If error variance varies with X, the simple homoskedastic standard-error formula need not be valid.

Precision of the Intercept: Where Is $X = 0$?

Under conditional mean zero and homoskedastic errors that are uncorrelated across observations conditional on $\mathbf{X}$,

$$\text{Var}(\widehat{\beta}_0 \mid \mathbf{X}) = \sigma_u^2 \left(\frac{1}{n} + \frac{\overline{X}^2}{TSS_X} \right).$$

Replace the unknown error variance by $SER^2 = SSR/(n-2)$ to estimate its standard error:

$$\boxed{SE(\widehat{\beta}_0) = SER \sqrt{\frac{1}{n} + \frac{\overline{X}^2}{TSS_X}}}.$$

- The intercept is the fitted outcome at $X = 0$.
- Holding n, error variance, and TSS_X fixed, the intercept is less precise when the sample's X values are centered farther from zero.
- If $\overline{X} = 0$, then $SE(\widehat{\beta}_0) = SER/\sqrt{n}$.
- A small standard error does not make an extrapolation substantively meaningful: check whether $X = 0$ is relevant to the population.

Use the standard error for the coefficient being studied; the intercept and slope generally have different standard errors and different units.

OLS in Large Samples: Consistency and Normal Approximation

Divide the numerator and denominator of the estimation error by n:

$$\widehat{\beta}_1 - \beta_1 = \frac{n^{-1}\sum_i (X_i - \overline{X})u_i}{n^{-1}\sum_i (X_i - \overline{X})^2}.$$

1. By the **Law of Large Numbers**, the numerator converges to $\text{Cov}(X, u) = 0$ under conditional mean zero.
2. The denominator converges to $\text{Var}(X) > 0$.
3. Therefore $\widehat{\beta}_1 \xrightarrow{p} \beta_1$: OLS is **consistent**.

Under the lecture's i.i.d. and moment assumptions, the **Central Limit Theorem** also gives asymptotic normality. With homoskedasticity,

$$\sqrt{n}(\widehat{\beta}_1 - \beta_1) \xrightarrow{d} N\left(0, \frac{\sigma_u^2}{\text{Var}(X)}\right).$$

Dividing $\widehat{\beta}_1 - \beta_1$ by an appropriate estimated standard error gives an approximately standard normal statistic in large samples. This parallels inference for $\overline{Y}$.

Lecture 1: Questions You Should Be Able to Answer

1. **Joint, marginal, or conditional?** In the sunny-day table, explain why $.55$, $.60$, and $.6875$ answer different questions.
2. **Why not reverse the condition?** Explain why $\Pr(Y = 1 \mid X = 1)$ and $\Pr(X = 1 \mid Y = 1)$ have different denominators.
3. **Which average?** Recover $\mathbb{E}[Y] = .60$ by weighting the two conditional means.
4. **What varies across samples?** Distinguish μ_Y, Y_i, $\overline{Y}$, and $\overline{y}$.
5. **Which theorem?** Use the Law of Large Numbers to explain consistency and the Central Limit Theorem to explain a normal approximation.
6. **Which implication is valid?** Independence implies zero covariance; the converse is false.

Key check

If n increases from 25 to 100 with the same population variance, the SD of the sample mean halves. The population SD of individual observations stays the same.

Lecture 2: An Inference Checklist

1. Identify the parameter: one mean or a difference of means.
2. State H_0, H_1, and α; determine whether the test is one- or two-sided.
3. Determine whether σ_Y is known. If not, estimate it with s_Y.
4. Calculate the **standard error of the estimator**, not just the SD of the observations.
5. Compute the observed statistic and choose a justified reference distribution.
6. Calculate the p-value or compare with a critical value; interpret the decision.
7. Construct a confidence interval and check consistency with the corresponding test.

Quick check

If $\bar{y} = 52$, $\mu_0 = 50$, $s_Y = 10$, and $n = 100$, then $SE(\overline{Y}) = 1$, $t_{\text{obs}} = 2$, and the two-sided normal-approximation p-value is about $.0455$. The 95% interval is $[50.04, 53.96]$.

Lecture 3: An OLS Checklist

1. Compute $\bar{x}, \bar{y}$, squared deviations of x, and cross-products.
2. Obtain $\widehat{b}_1 = s_{XY}/s_X^2$ and $\widehat{b}_0 = \bar{y} - \widehat{b}_1 \bar{x}$.
3. Calculate $\widehat{y}_i$ and residuals $y_i - \widehat{y}_i$. Check the two sample FOCs.
4. Compute SSR, TSS_Y, ESS, R^2, and SER; interpret their units and meanings.
5. State the assumptions needed for unbiasedness, consistency, and the standard-error formula.
6. Explain whether the slope has a causal interpretation; identify plausible omitted variables.

Do not confuse these statements

$\sum_i \widehat{u}_i = 0$ is an OLS identity.
$\mathbb{E}[u_i \mid X_i] = 0$ is a population assumption.
A high R^2 describes fit. A small standard error describes precision.
Neither one alone establishes causality.